



Data Structures and Algorithms -3 (25CS2103E)

Product Search and Recommendation System for Shopping

BY TEAM 4

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Abstract

The Product Search and Recommendation System for Shopping is designed to provide a smarter and more efficient way for users to search and discover products on an online shopping platform. With the rapid growth of e-commerce, customers often find it difficult to locate the right product due to incorrect spellings, incomplete product names, or a large number of similar products. The proposed system overcomes these challenges by enabling users to search using product names, categories, brands, keywords, or descriptive queries. It retrieves the most relevant products from a structured product repository and presents them in an organized and user-friendly manner, making the search process faster and more accurate.

In addition to efficient product search, the system recommends similar or related products based on their category, brand, features, and user preferences. It also allows users to apply filters such as price range, brand, ratings, and availability to narrow down their search results and make informed purchasing decisions. By combining intelligent search with personalized recommendations, the system enhances the overall shopping experience, reduces the time spent searching for products, and helps users easily discover products that best match their needs. The proposed solution is scalable, reliable, and suitable for modern e-commerce applications that manage large product collections.

Introduction

Online shopping has become an important part of everyday life, with e-commerce platforms offering thousands of products in different categories, brands, and price ranges. As the number of products increases, users may find it difficult and time-consuming to locate the exact product they need. Searching becomes even more difficult when users enter incomplete product names, spelling mistakes, or describe their requirements instead of knowing the exact product name.

The **Product Search and Recommendation System for Shopping** aims to provide a simple and intelligent solution to these problems. The system allows users to search for products using keywords or descriptive queries and provides features such as spelling correction, auto-complete, and product filtering. It also supports intent-based searches such as *“laptop for gaming under ₹80,000”* by identifying the user's basic requirements and finding suitable products. In addition, the system recommends relevant products and provides reasons for the recommendations, helping users compare options and make better purchasing decisions.

Literature Survey

Existing e-commerce platforms provide product search and recommendation features to help users find products from large collections. Traditional search systems mainly match user-entered keywords with product names, descriptions, and categories. Modern platforms also provide features such as **filters, auto-complete, spelling correction, and product recommendations** to improve the shopping experience. These features help users narrow down large product collections and find suitable products more easily. However, users may still receive irrelevant results when they enter spelling mistakes, incomplete queries, or describe their requirements instead of using an exact product name.

The proposed system builds upon these existing approaches by combining **smart product search and explainable recommendations** in a single system. It focuses on handling spelling errors, providing search suggestions, applying filters based on price, brand, and category, and understanding requirement-based queries such as *“gaming laptop under ₹80,000.”* The system identifies important details from the user's query, searches the available product collection, and ranks products according to how closely they match the requirements. It also explains why each product is recommended, helping users understand the results, compare suitable options, and make better purchasing decisions.

Problem Statement and Objectives

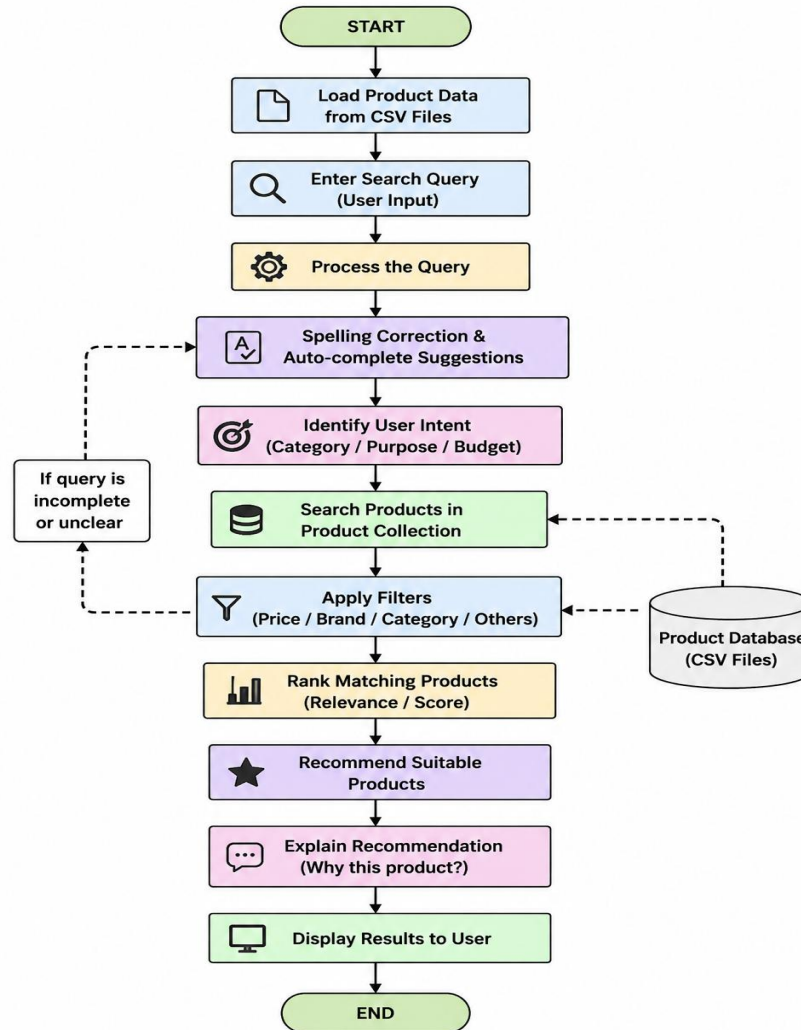
Problem Statement

Modern e-commerce platforms contain millions of products, making it difficult for customers to quickly find the items they need. Traditional search systems often fail when users enter incomplete product names, spelling mistakes, or search using natural language such as "laptop for programming under ₹60,000." As a result, users receive irrelevant search results, spend more time searching, and may fail to discover suitable alternatives. There is a need for an intelligent product search and recommendation system that understands user intent, retrieves relevant products efficiently, and recommends similar products to improve the overall shopping experience.

Objectives

- To provide fast and efficient product searching.
- To handle common spelling mistakes in user queries.
- To provide auto-complete suggestions while searching.
- To filter products based on price, brand, and category.
- To understand simple user requirements from descriptive search queries.
- To identify and rank products that best match the user's requirements.
- To recommend suitable and related products.
- To explain why each product is recommended.
- To improve the overall product discovery and shopping experience.

Flow Chart



Results

The proposed system provides faster and more relevant product search results by processing user queries and matching them with products stored in the product repository. It can handle spelling mistakes, provide auto-complete suggestions, and apply filters such as **price, brand, and category**. The system also supports requirement-based searches such as “**gaming laptop under ₹80,000**” by identifying the product type, purpose, and budget. Based on these requirements, the system searches the available products and ranks the most suitable results.

The system displays important product details such as **product name, brand, category, price, rating, and match score**. It also recommends similar and relevant products based on the user's search requirements. Along with the recommendations, the system provides a simple explanation of why each product was selected, such as matching the user's budget, category, features, or rating. The expected result is a simple, efficient, and user-friendly shopping search system that reduces search time and helps users compare and select suitable products more easily.

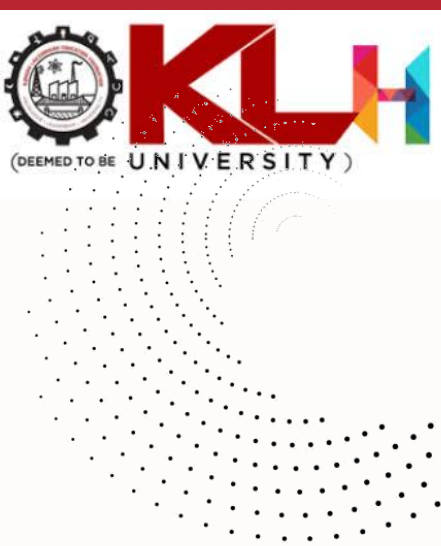
Hardware and Software requirements

Hardware Requirements

- **Processor:** Intel Core i3 or equivalent and above
- **RAM:** Minimum 4 GB
- **Storage:** Minimum 10 GB free space
- **Display:** Standard monitor
- **Input Devices:** Keyboard and mouse

Software Requirements

- **Programming Language:** Java
- **JDK:** JDK 17 or above
- **IDE:** Visual Studio Code (VS Code)
- **Java Extension:** Extension Pack for Java
- **User Interface:** Java Swing
- **Data Storage:** CSV / Text Files
- **Operating System:** Windows / Linux
- **Version Control:** GitHub



Conclusion

The **Product Search and Recommendation System for Shopping** demonstrates how a simple product repository can be transformed into a more useful and intelligent shopping assistant. Instead of relying only on exact product names, the system considers what the user is actually trying to find and helps narrow down the available choices.

The project also shows how different algorithmic techniques can be applied to solve practical problems in e-commerce. Features such as search assistance, requirement-based matching, product ranking, and recommendation explanations make the system more useful from a user's perspective. Since the system is implemented using **Java and file handling**, it remains simple to develop and understand while providing a strong foundation for future improvements.

Overall, the project provides a practical combination of **search, decision support, and recommendation**, with the goal of making product discovery quicker, clearer, and more convenient for users.

Future Scope

- The system can be further developed to provide a more advanced and personalized shopping experience. Future improvements can include:
 - **Voice-based search** for users who prefer speaking instead of typing.
 - **Image-based search** to find products using an uploaded product image.
 - **Personalized recommendations** based on user search and purchase history.
 - **Advanced AI-based intent understanding** for more complex natural-language queries.
 - **Real-time price comparison** across different shopping platforms.
 - **Price-drop notifications** when the price of a selected product decreases.
 - **Online database integration** to handle a much larger and continuously changing product collection.
 - **Mobile application** so users can access the system from smartphones.
 - **User feedback integration** to improve future recommendations.
 - **Multilingual search** to allow users to search for products in different languages.

These enhancements can transform the current system from a basic intelligent search application into a more complete **personalized shopping assistant**.

Thank you